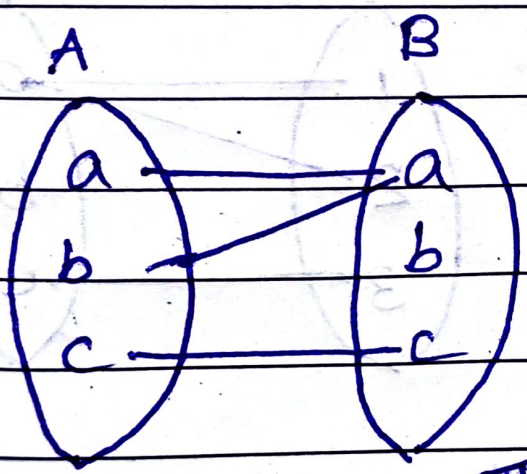
Functions

Let A and B be 2 non-empty sets. The relation which assign the element of A to the unique element of B is called function from A to B.
written as $f: A \to B$.

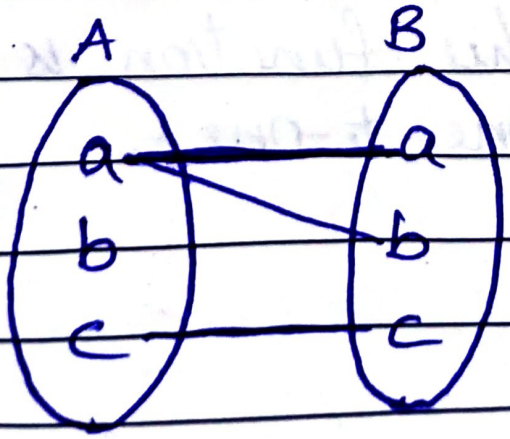


$f = \{(a,a), (b,a), (c,c)\}$

Domain $(f) = \{a, b, c\}$

Range $(f) = \{a, c\}$

This is an example of function.

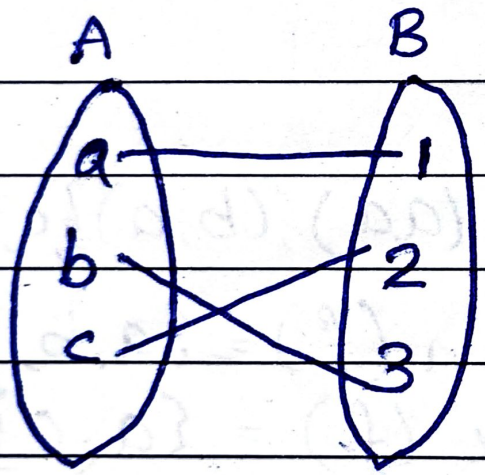


But this is not a function since element of A is assigned to two elements of B.

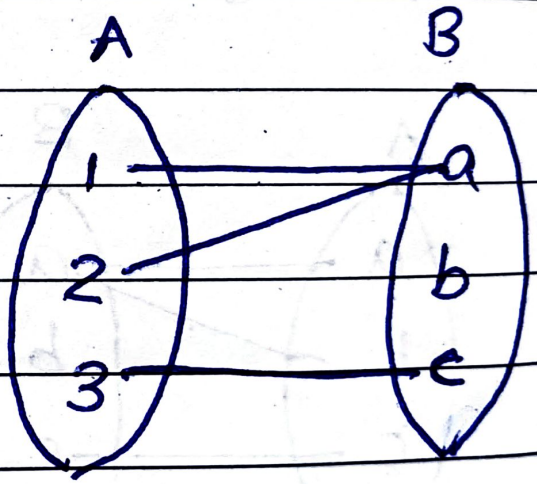
Types of function

1. One-to-One (1-1) or Injective

A function $f: A \to B$ is said to be 1-1 if different elements have distinct images.



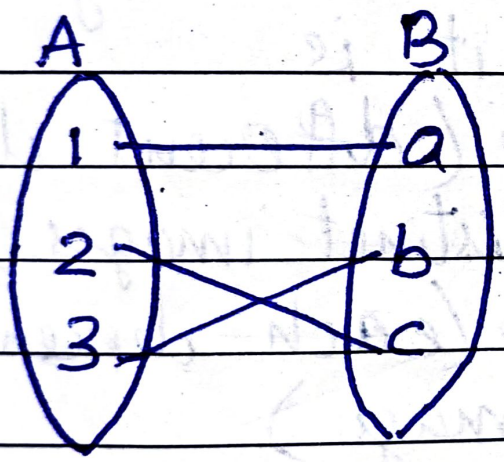
This is one-to-one function



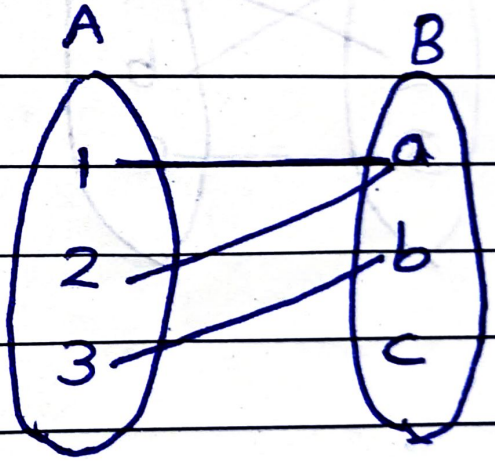
Here, a is the image of 2 elements. So this function is not one-to-one.

2. Onto or Surjective

A function $f: A \to B$ is said to be onto function if each element of B is the image of some element of A then A is onto.



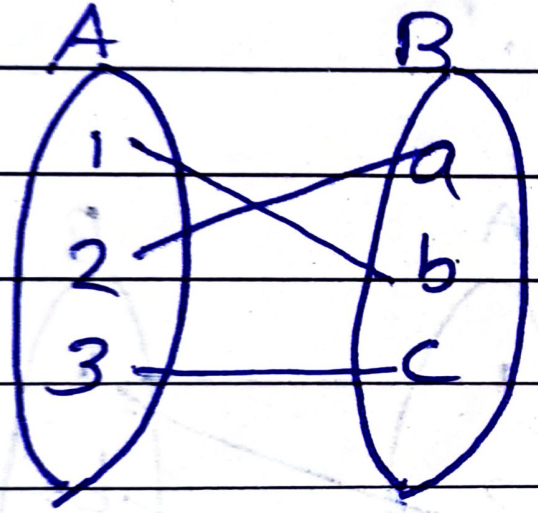
This is onto function



Here c is not assigned to any element of A So this function is not onto.

Bijective

A function $f: A \to B$ is said to be bijective if it is both one-one and onto.



This function is bijective since it is one-one (different elements have distinct images) and onto (each element in A has image)